to be on the line l. The distances of villages A and B from the foot of the perpendiculars P and Q on the line l are 3 km and 5 km respectively and the distance between P and Q is 6 km. How to calculate **the smallest length** of cable required to connect the two villages (or the roads that connect the villages as well as the power station) from the power station and to find the equations of the cable lines (or roads) that connect the power station to two villages.

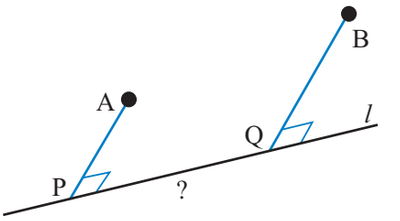
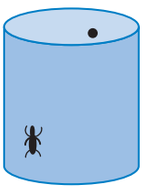


Figure 6.1

Real life situation 6.3:



Consider a hollow cylindrical vessel, with circumference 24 cm and height 10 cm. An ant is located on the outside of vessel 4 cm from the bottom. There is a drop of honey at the diagrammatically opposite inside of the vessel, 3 cm from the top. What is the shortest distance the ant would need to crawl to get the honey? What is the equation of the path traced out by the ant. Here is a picture that illustrates the position of the ant and the honey.

Figure 6.2

Real life situation 6.4: The quantity demanded of a certain type of Compact Disk is 22,000 units when a unit price is ₹ 8. The customer will not buy the disk, at a unit price of ₹ 30 or higher. On the other side the manufacturer will not market any disk if the price is ₹ 6 or lower. However, if the price is ₹ 14 the manufacturer can supply 24,000 units. Assume that the quantity demanded and quantity supplied are linearly proportional to the price. How to find (i) the demand equation (ii) supply equation (iii) the market equilibrium quantity and price. (iv) The quantity of demand and supply when the price is ₹ 10.

The equation of the straight line for each of the problems stated above, not only solves the specific case of solutions but also helps us get many information through it. Later, in this chapter, let us try to solve these types of problems by using the concepts of straight lines. In order to understand the straight line, we need to get acquainted with some of its basic concepts. Let us discuss those in detail

Learning Objective

On completion of this chapter, the students expected to know

- the equation of a line in different forms
- whether two given lines are parallel or perpendicular;
- the distance of a given point from a given line and between two parallel lines,
- the family of straight lines for a given condition
- the equation of pair of straight lines, angle between them and angle bisectors

6.2 Locus of a point

Definition 6.1

A **point** is an exact position or location on a plane surface.

